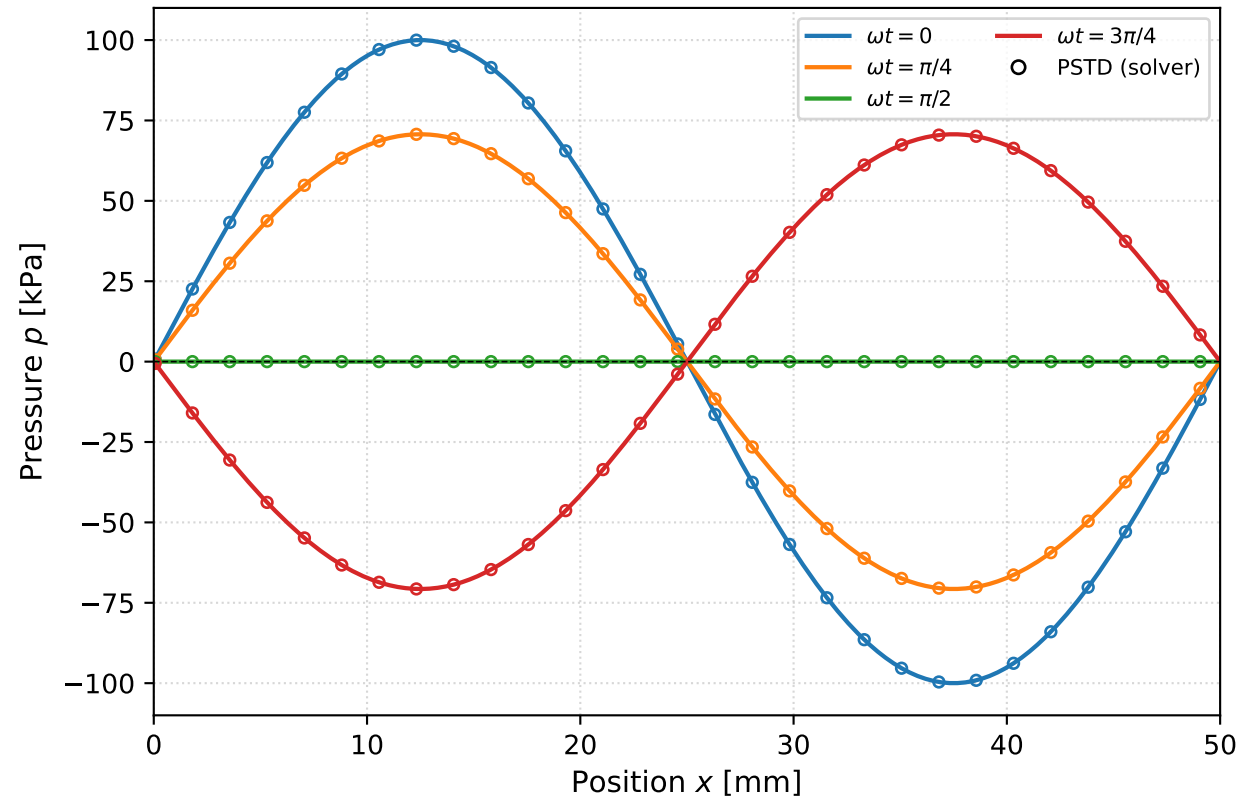


Figure 1.1 — Wave-equation solver validation in a water column ($c_0 = 1481$ m/s, $\rho_0 = 998$ kg/m³)

(a) Standing wave — PSTD (transparent boundary) vs analytic
 $p = p_0 \sin(kx) \cos(\omega t)$ (max error 0.00%)



(b) Travelling pulse — FDTD vs d'Alembert (Theorem 1.2)
 split into $\pm c_0 t$ pulses at $c_0 = 1481$ m/s; 3.7% shape error is FDTD numerical dispersion

